

James Lin

Senior Full Stack Engineer | ex-AWS | React | TypeScript | Node | Python | Go | AWS

Passionate Senior Software Engineer with over 10 years of experience building scalable, user-centric applications across industries like healthcare, finance, and e-commerce. Proficient in **TypeScript (React, Node.js)**, **Python (Django, Flask)**, and AI-driven development. I specialize in crafting robust backends and integrating intelligent solutions to enhance performance and functionality. A fast learner and problem-solver, I thrive on tackling complex challenges with cutting-edge technologies in both startup and enterprise environments. Eager to join a forward-thinking team, I'm ready to drive innovation and deliver impactful software solutions.

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New York, NY 

PROFESSIONAL EXPERIENCE

Tactiq – Richmond, VA (Remote)

Senior Software Engineer

10/2021 - 3/2025

- Led a team of backend and frontend engineers in developing and maintaining full-stack web applications (**GlobalWorx**), overseeing the entire software development lifecycle.
- Designed and developed a **RESTful API** in **Python (Django)** and **PostgreSQL**, serving over 1,000 daily users and handling high request volumes.
- Built a **serverless API** using **AWS API Gateway**, **Lambda**, **PostgreSQL**, and **DynamoDB**, enabling seamless client integration with our **SaaS platform**.
- Developed a scalable, event-driven ETL pipeline with **Go/Python**, **Kubernetes**, **PostgreSQL**, and **Kafka**, processing thousands of records per minute.
- Implemented a **CI/CD framework** using **Docker**, **Terraform**, and **GitHub Actions**, streamlining microservice deployments and reducing time from development to production.
- Created **data dashboards and automated reports** with **Looker**, delivering actionable insights to internal and external stakeholders.
- Developed **SQL models** to enable self-serve reporting for non-technical stakeholders, improving data accessibility and decision-making.
- Automated administrative and reporting tasks using **Python (Pandas, NumPy)**, saving significant labor hours and reducing errors.
- Conducted **ad-hoc data analysis** using **SQL** and **MongoDB**, tracking KPIs to drive data-informed decisions.
- Led critical development efforts in a **large-scale invoicing project (Acuity)**, enhancing performance, reliability, and maintainability.
- Designed and implemented a **custom access authorization system** for invoicing, resolving key bugs and improving maintainability.
- Spearheaded a **major system refactoring**, accelerating development and improving scalability.
- Owned critical financial functionalities, including **tax calculation and invoice generation**, ensuring accuracy and compliance.
- Collaborated with stakeholders to gather and structure requirements, driving complex projects from concept to execution.

Amazon Web Services – Seattle, WA (Remote)

Software Development Engineer II

08/2019 - 09/2021

- **Led** the development of an e-learning platform that enhances **AWS** learning with rich video experiences.
- Built a client-facing application using **Next.js** with high streaming capabilities.
- Created a video management portal for users to upload videos and monitored the video processing pipeline.
- Directed a team of three in building a serverless video processing pipeline using **AWS CDK** and **TypeScript**.
- Worked with Applied Scientists to convert video-processing **Python** scripts into **AWS Lambda** functions.
- Wrote unit and integration tests for the video processing pipeline using **Jest** and **Pytest**.
- Created a custom video player component with **React**, integrating **AI** capabilities like transcripts, deep dives and OCR.
- Improved accessibility and internationalization by following **WCAG** guidelines and providing translatable text content.

Talkspace – New York, NY (Onsite)

Software Engineer

01/2015 - 07/2019

- **Spearheaded** the development of telehealth web and mobile applications with AI capabilities from scratch.
- Implemented responsive and pixel-perfect UIs using **React** with **TypeScript** and **React Native (Expo)**.
- Utilized **Node.js**, and **Flask** framework for building micro-service backend infrastructure, establishing communication between the **micro-services** using **RabbitMQ**.
- Successfully implemented **multi-lingual** real-time text and video chat application with AI integration using **WebSocket**, and **WebRTC**.
- Participated in building a **multi-lingual** real-time communication and streaming platform creating a highly scalable and responsive web application using **Next.js**, **Node.js**, **GraphQL**, and **PostgreSQL** with **TypeScript**.
- Utilized technologies such as **Docker** and **Kubernetes** to containerize microservices, enabling seamless deployment and management of services in a cloud-native environment.
- Implemented data caching strategies using **Redis**, reducing page load times by 20% and enhancing overall website performance.
- Utilized an **IaC** (Infrastructure as Code) tool-**Terraform** for managing and provisioning application infrastructure.
- Managed **AWS** services like **EC2**, **ECS**, **Lambda**, **IAM**, **SQS**, **RDS**, **DynamoDB**, **Cognito**, **Amplify**, **Media Live**, and **Media Converter**.
- Configured and managed **AWS Elastic Load Balancers** (ELB) to ensure high availability and fault tolerance for web applications.
- Utilized built-in **SQLite** database to store data related to the hospital, such as doctor and patient information under **HIPAA** compliance.
- Boosted **SEO** with **SSR**, **SSG**, metadata, and image optimization, by implementing responsive, semantic, and mobile-friendly pages with minimized loading time.
- Implemented unit tests using **Pytest** for features to ensure stability.
- Designed **GraphQL** schemas and defined specific queries, optimized data fetching while minimizing over-fetching or under-fetching.
- Developed and implemented robust authentication and authorization mechanisms using **Firebase**, ensuring secure access control and user management for web and mobile applications.
- Developed and deployed microservices to manage various functional components of the application, utilizing **Docker** and **Kubernetes** for container orchestration.
- Actively participated in Agile ceremonies, including sprint planning, daily stand-ups, and retrospectives while contributing to continuous integration and delivery (**CI/CD**) pipelines using **GitHub Actions** and **Git**.
- Implemented data visualization for patient health metrics using **D3.js** and **Canvas**.
- Provided **mentorship** to junior developers, conducted code reviews and pair programming, and ensured adherence to best coding practices and standards.

- Actively collaborated with **AI** engineers and used **Python** and **SQL** to transform and clean data for developing disease identification features using visual data such as images or videos.
- Closely worked with PM and CTO to gather and transfer the client's requirements into technical directions.
- Collaborated with cross-functional teams to design and deploy **Python**-based solutions, ensuring alignment with business requirements.

EDUCATION

Bachelor's Degree in Computer Science, Columbia University, New York **2009 – 2013**

Master's Degree in Computer Science, Columbia University, New York **2013 – 2014**

ADDITIONAL SKILLS

- C/C++, Java, C#, Golang, Rust, Ruby on Rails, Scala, MySQL, MongoDB
- React Native, Angular, Vue, Svelte, Bootstrap, CSS, MUI, Mobx, Zustand, Material UI, Tailwind CSS, Ant Design
- Numpy, FAISS, Langchain, Spark, Celery, Neo4j, SQLAlchemy, Selenium, Elastic Search, Kibana
- Documentation, Out-of-box thinking, Forward thinking, Active listener, Prioritizing tasks, Team play
- Technical Leadership
- Jenkins, Kafka, Cypress, Playwright, Enzyme